RESEARCH ASSIGNMENT 1

SECTION A : Database Fundamentals

1. What are the main types of databases?

- Rational Databases (RDBMS): store data in tables with rows and columns. They use SQL for data manipulation and have strict, predefined schemas eg MySQL, Oracle, and Microsoft SQL server.
- Non-Rational Databases (NoSQL): Designed for large volumes of unstructured or semi-structured data. They offer flexible schemas and are often used for applications like social media and IoT data eg include document, key-value, columnar, and graph databases.

2. What is a Rational Database Management System (RDBMS)?

- An RDBMS is a type of database management system that stores data in a structured format, using tables with rows and columns.
- It establishes relationships between different tables using keys and allow users to access, manage and update the data through a query language, typically Structured Query language (SQL)

3. What is a primary key and a foreign key in a database?

- A primary key is a column or a set of column that uniquely identifies each row in a table. No two rows can have the same primary key value and it cannot be null e.g studen_id
- A foreign key is a column or set of columns in one table that refers to the primary key in another table. It is used to establish and enforce a link between the two tables.

4. What is database normalization and why is it important.

- Normalization is the process of organizing data in a database to reduce redudancy and improve data integrity. This involves structuring columns and tables to ensure data dependencies are as logical as possible.
- Importance: It makes the database more efficient by avoiding redudant data, which save storage space and makes updates more consistent. It also helps in designing a flexible and well structured database that is easier to maintain and scale.

5. What is a database schema?

A database schema is the blueprint of the underlying structure of a database. It defines how the data is organized, including the tables, the relationship between them and the data types for each column.

6. Differentiate between structured, semi-structured and unstructured data.

- Structured data: Highly organized and follows a fixed format with predefined relationships, like data in a relational database table (tables, rows and columns)
- Semi-structured data: Does not conform to rigid, relational model but has some organizational properties e.g JSON or XML files, where tags or keys describe the data.
- Unstructured data: Lacks a predefined organization, such as text documents, videos, audio files and images.

7. What is the difference between a Fact Table and a Dimension Table in a data warehouse.

- Fact Table: A central table that contains quantitative measures (facts) and foreign key linking to dimension tables. It holds the business metrics and numerical data
- Dimension Table: A table that contains descriptive attributes (dimensions) related to the fact table e.g

a customer dimension table might contain customer name, address and cities.

8. What is a data model and why is it important in database design?

- A data model is an abstract representation of a database's structure. It defines the data elements, their relationships and the rules governing them.
- Importance: It is crucial in database design as it provides a blueprint that helps in visualizing and organizing data before implementation, ensuring that the database will meet the required business needs and constraints.

9. Explain the difference between a database, a data warehouse and a data lake.

- Database: Stores current, operational data
- Data Warehouse: Stores historical, structured data for analysis.
- Data lake: Stores all types of raw data (structured, semi, unstructured).

10. What is a data mart and how does it it differ from a data warehouse.

A data mart is a subset of a data warehouse that is focused on a specific business function or department such as finance, sales or marketing. A data warehouse is a large centralized repository that stores integrated data from across the entire organization to provide a holistic view for enterprice-level decision-making.

- The key difference is the scope, size and purpose with data marts being more specific and faster to implement for departmental analysis, while data warehouse are broader and more complex for company-wide insights.

SECTION B: SQL and Data Processiong.

1. What is a query language, and why is SQL the most commonly used?

A query language is a specialized computer language used to retrieve and manage data from databases. SQL (Structured Query language) is the most commonly used due to its standardization, widespread adoption across various relational database management systems (RDBMS) ease of learning with its English-like-syntax and its powerful capabilities for data manipulation, definition and control.

2. What are indexes in databases and how do they improve performance?

Indexes are special lookup tables that the database search engine can use to speed up data retrieval. They work by creating a sorted structure of one or more columns in a table the database to quickly locate specific rows without scanning the entire table. This significantly improves the performance of SELECT statements, WHERE clauses, JOIN operations and ORDER BY clauses.

3. What are transactions in databases and what are the ACID properties?

A transaction is a single logical unit of work performed on a database. It can consist of one or more operations but it is treated as a complete, indivisible unit. The ACID properties ensure the reliability of transactions:

- Atomicity: All operations within a transaction either complete successfully or none of them do.
- Consistency: A transaction brings the database from one valid state to another valid state.
- Isolation: Concurrent transaction execute independently without interfering with each other.
- Durability: Once a transaction is committed, it changes are permanent and survive system failures.

4. What is a database engine and how does it impact performance?

A database engine (storage engine) is the underlying software component that manages how data is stored, retrieved and updated in a database. It significantly impacts performance through its implementation of indexing, locking mechanisms, caching strategies and query optimization algorithms. Different engines (e.g InnoDB, MyISAM) offer varying features and performance characteristics suited for different workloads.

5. What are views, stored procedures and triggers in SQL?

- Views: Virtual tables based on the result-set of a SQL query. They do not store data themselves but provide a simplified and secure way to access specific data from one or more underlying tables.
- Stored Procedures: Pre-compiled SQL code blocks stored within the database. They can accept parameters, execute complex logic and return results, offering reusability, improved performance and enhanced security.
- Triggers: Special stored procedures that automatically execute in response to specific events (e.g INSERT / UPDATE, DELETE) on a table. They are used to

enforce business rules, maintain data intergrity and automate task.

6. Differentiate between ETL (extract, transform, load) and ELT (Extract, load, Transform).

• ETL: Data is extracted from source systems, then transformed (cleaned, standardized, aggregated) in a staging area and finally loaded into a target data warehouse and data mart.

• ELT: Data is extracted from source systems, then directly loaded into a target data warehouse or data lake, and finally transformed within the target system using its processing capabilities. ELT is often preferred for large datasets and cloud-based data warehouse.

7. Differentiate between batch processing and stream processing in data pipelines.

• Batch processing: Process data in large, predefined batches at scheduled intervals. It is suitable for historical data analysis and scenarios where real-time insights are not critical.

• Stream Processing: Processes data continuously as it arrives, in real-time or near real-time. It is ideal for applications requiring immediate insights, such as fraud detection or real-time dashboards.

8. Explain what a join is in SQL and list different types of joins with examples.

A JOIN in SQL combines rows from two or more tables based on a related column between them.

- INNER JOIN: Returns rows where there is a match in both tables.

```
SELECT * FROM orders AS A INNER JOIN customers AS B
ON. A.customer_id = B.customer_id;
```

- LEFT JOIN: Returns all rows from the left table and matching rows from the right table. If no match, NULL are returned for right table columns.

```
SELECT * FROM customers AS A
LEFT JOIN orders AS B
ON. A.customer_id = B.customer_id;
```

- RIGHT JOIN: Returns all rows from the right table and matching rows from the left table. If no match, NULLS are returned for left table columns.

```
SELECT * FROM orders AS A
RIGHT JOIN customers AS B
ON A.customer_id = B.customer_id;
```

• FULL JOIN: Returns all rows when there is a match in one of the tables. Returns nulls for columns where there is no match in the other table.

```
SELECT* from customers AS A
FULL JOIN orders AS B
ON A.customers-id = B.customers-id;
```

9. What is referential integrity and why is it important in relational databases?

Referential integrity is a database concept that ensures relationships between tables remain consistent. It is enforced using foreign keys, which link a column in one table (the child table) to the primary key in another table (the parent table). It is important because it prevents orphaned records, ensures data consistency and maintains the validity of relationship between data.

10. How does data redundancy affect database performance and storage.

Data redundancy occurs when the same data is stored in multiple places within a database.

• Performance: It can negatively impact performance by increasing storage space requirements, making updates more complex (as multiple copies need to be updated) and potentially leading to data inconsistencies if not managed properly.

- Storage: It increases the overall storage footprint of the database, as duplicate data consumes unnecessary space.

Section C: Data Management and Analytics Concepts.

21. How does cloud database management differ from on-premise database?

- Cloud databases: are hosted and managed by a third-party provider (e.g AWS, Azure, Google Cloud). They are scalable, cost-efficient and accessible from anywhere via the internet. Maintenance tasks such as update, backups and scaling are automated by the provider.
- On-premise databases are installed locally within an organization's own servers. They offer more control and security customization but required in-house management, hardware costs and manual maintenance.

Cloud = flexible and managed

On-Premise = Controlled but resource-intensive

22. What is data governance and why is it important in data management?

Data governance is set of policies, role and procedures that define how data is managed, accessed and protected across an organization.

It ensures that data is accurate, consistent, secure and compliant with regulations (like POPIA or GDPR)

Importance:

=> Maintains trust in data used for decisions.

=> Protects sensitive information.

=> Ensures accountability and compliance

23. What is data integrity and how can it be maintained.

Data integrity means that data remains accurate, complete and reliable throughout its lifecycle. It ensures that data isn't altered or corrupted accidentally.

Maintained by:

- Using primary and foreign key constraints.
- Applying validation rules and input controls.
- Implementing backups and access controls.
- Using transaction management (ACID properties) to keep data consistent.

24. What is data quality and why is it critical for analytics?

Data quality measures how well data meets the needs of its inted use, considering accuracy, completeness, timeliness, consistency and validity.

Why it Matters:

Poor-quality data leads to incorrect insights, bad bussiness decisions and loss of trust in analytics.

High quality data ensures reliable reporting, predictive accuracy and effective decision making.

25. Explain the role of data analysis in managing and analyzing database information.

A data analysis plays a key role in bridging data and decision making.

Responsibilities include:

1. Collecting and cleaning raw data from databases.
2. Writing SQL queries to extract and combine data
3. Performing analysis to uncover trends and pattern
4. Creating dashboards and report for stakeholders.
5. Ensuring data accuracy and consistancy.

Data Analysis transform data into actionable insights that help guide business strategies.

26. What are the key responsibilities of a Database Administrator (DBA)?

A DBA is responsible for the technical management of databases.

Key duties include:

- Installing and configuring database systems.
- Managing user access and security permissions.
- Monitoring performance and optimizing queries.
- Performing backups and recovery
- Applying patches and upgrades.
- Ensuring data availability, consistency and reliability.

27. What are the main steps involved in designing a data pipeline.

A data pipeline automates the flow of data from source to destination.

Main steps:

1. Identify data sources (databases, APIs, files etc)
2. Extract data from these sources
3. Transform data (cleaning, formatting, aggregating)
4. Load data into a destination (data warehouse, lake or dashboard).
5. Orchestrate and schedule the process for automation.
6. Monitor and Maintaine for performance and accuracy.

28. What are some common challenges in managing large scale databases?

Managing large databases can be difficult due to:

⇒ Performance issues (slow queries, indexing problems)

⇒ Data consistency and integrity across distributed systems.

⇒ Storage and scalability demands as data grows.

⇒ Backup and disaster recovery complexity.

⇒ Security risks and access control.

⇒ Cost of infrastructure and maintenance.

29. What are some popular database platform and their use cases?

Platform	Type	Use Case
MySQL	Relational	Web applications and small-to medium systems.
PostgreSQL	Relational	Complex queries, advanced analytics and enterprise applicat
Oracle	Relational	Large-scale enterprise systems with high transactional volumes.
SQL Server	Relational	Microsoft ecosystem applications and BI tools.
Snowflake	Cloud data ware-house.	Big data analytics, cloud-native platforms.
MongoDB	NoSQL (Doc)	Unstructured data, real-time apps, IoT

30. What are the main data storage formats used in analytics?

Format	Type	Description / Use case
CSV (comma-separated values)	Structured	Simple tabular text format used for exports / imports.
JSON (JavaScript Object Notation)	Semi-struct	Hierarchical data exchange, APIs, NoSQL databases.
Parquet	Columnar	Highly efficient for big data analytics.
Avro	Row-based binary	Ideal for streaming pipelines and schema evolution.
ORC	Columnar	Used in Hive and Hadoop systems for compressed storage